Final Exam

● Graded

Student

Yifei Pan

Total Points

45 / 45 pts

Question 1

Overall Grades **45** / 45 pts

✔ **– 0 pts** See the details on each problem.

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Yifei Pan	YP167	潘逸飞

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 潘逸飞

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	5	5	5	45

Problem 1. [5'] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right]$$

$$\sim \begin{matrix} \\ \\ -2R_1 \end{matrix} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$$\sim \begin{matrix} \\ \\ -\frac{3}{2}R_2 \end{matrix} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & \frac{1}{2} & -\frac{3}{2} \end{array}\right]$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$$

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

1). Col(A) = span $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix} \right\}$

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 2 & 1 \end{bmatrix} \sim \begin{matrix} \\ \\ +R_1 \\ \\ \end{matrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \\ 1 & 2 & 1 \end{bmatrix} \sim \begin{matrix} \\ \\ \\ -R_1 \end{matrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ So, A basis of col(A) is $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

2). row(A) = span $\left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}^T, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix}^T, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}^T, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}^T \right\}$

$$\begin{bmatrix} 2 & -5 & 0 & 1 \\ 4 & 0 & 1 & 3 \\ 2 & 5 & 1 & 2 \end{bmatrix} \begin{matrix} \\ -2R_1 \\ -R_1 \end{matrix} \sim \begin{bmatrix} 2 & -5 & 0 & 1 \\ 0 & 10 & 1 & 1 \\ 0 & 10 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 0 & 1 \\ 0 & 10 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So, a basis of col(A) is $\left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}^T, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix}^T \right\}$

For null space X,

3). $AX = 0$

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$$\sim \begin{matrix} \\ +R_1 \\ -R_1 \\ \\ \end{matrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} s$$

So, a basis of null(A) is $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$$A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

Take $\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$ as X, find X^{-1}

$$\left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right] \sim \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\-1&-1&-1&0&0&1\end{array}\right] \sim +R_2\left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&-1&-1&0&1&1\end{array}\right]$$

$$\sim +R_2\left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&-1&1&1&1\end{array}\right] \sim \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right]$$

$$X^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

$$A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$$

Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

① $\det(A-\lambda I) = 0$

$$A-\lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$$

$$\det(A-\lambda I) = (1-\lambda)(-1)^6 \begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix}$$

$$= (1-\lambda)(1-\lambda)(2-\lambda) = 0$$

So, $\lambda_1 = 1$, whose $AM = 2$

$\lambda_2 = 2$, whose $AM = 1$

For $\lambda_1 = 1$,

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\langle x, y, z \rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} t$$

So one eigenvector is $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, call it v_1

For $\lambda_2 = 2$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \sim \begin{matrix} \\ \\ -2R_1 \end{matrix} \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & -6 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 2 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$ So, $\langle x, y, z \rangle = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix} s$

So one eigen vector is $\begin{bmatrix} -2 \\ -1 \\ 6 \end{bmatrix}$, call it v_3

$$Av_2 = v_1 + v_2$$

$$(A-I)v_2 = v_1,$$

We then have

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{vmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{vmatrix}$$

$$\langle x, y, z \rangle = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} t$$

Take $t = 0$, one v_2 is $\begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$

So, $J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is the jordan canonical form

with $P = \begin{bmatrix} 0 & -\frac{1}{2} & -2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix}$

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Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = X^T A X$

$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$

$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix}$

$= (3-\lambda)(-1)^2 \begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + 1\times(-1)^4 \begin{vmatrix} 0 & 3-\lambda \\ 1 & -1 \end{vmatrix}$

$= (3-\lambda)\left[(3-\lambda)(4-\lambda)-1\right] + \left(0-(3-\lambda)\right)$

$= (3-\lambda)\left[\lambda^2 - 7\lambda + 11 - 1\right]$

$= (3-\lambda)(\lambda-2)(\lambda-5)$

So, $\lambda_1 = 2$, $\lambda_2 = 3$, $\lambda_3 = 5$

For $\lambda_1 = 2$, $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \xrightarrow{-R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ So, $\langle x,y,z\rangle = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} s$

one eigenvector, $v_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

For $\lambda_2 = 3$ $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

So, $\langle x,y,z\rangle = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} s$.

one eigenvector, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

For $\lambda_3 = 5$,

$\begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \xrightarrow{R_3} \sim \begin{bmatrix} 0 & -2 & -1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$, So, $\langle x,y,z\rangle = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} s$

one eigenvector, $v_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

v_1, v_2, v_3 are orthogonal

because, $v_1 \cdot v_2 = 0$, $v_1 \cdot v_3 = 0$, $v_2 \cdot v_3 = 0$.

So, $P = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}$

since $X = PU$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{u}{\sqrt{3}} + \frac{v}{\sqrt{2}} + \frac{w}{\sqrt{6}} \\ \frac{u}{\sqrt{3}} + \frac{v}{\sqrt{2}} - \frac{w}{\sqrt{6}} \\ \frac{u}{\sqrt{3}} + \frac{2}{\sqrt{6}} w \end{bmatrix}$

$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$f = U^T D U$

$= 2u^2 + 3v^2 + 5w^2$

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Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = \begin{bmatrix} 0 \\ 2a \\ -2a \end{bmatrix} + \begin{bmatrix} 2b \\ -b \\ 2b \end{bmatrix} + \begin{bmatrix} -c \\ c \\ 0 \end{bmatrix} = \begin{bmatrix} 2b-c \\ 2a-b+c \\ -2a+2b \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$$

Then we have $\begin{cases} 2b-c=-3 \\ 2a-b+c=8 \\ -2a+2b=-8 \end{cases}$,

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & -3 \\ 0 & 2 & -1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \sim \xrightarrow{+R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \sim \xrightarrow{\div 2R_3} \left[\begin{array}{ccc|c} 0 & 0 & -3 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right], \text{ so we have } \begin{cases} a=3 \\ b=-1 \\ c=1 \end{cases}$$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Take, $u_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, $v_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$

$v_2 = v_1 - proj_{u_1}(v_1)$

$proj_{u_1}(v_1) = \frac{v_1 \cdot u_1}{u_1 \cdot u_1} \cdot u_1 = \frac{2}{2} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

So, $v_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

So, $u_1' = \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}$, $v_2' = \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}$

So an orthonormal basis of V is

$$\left\{\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\\frac{1}{-2\sqrt{3}}\end{bmatrix}\right\}$$

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Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

From Properties of determinants.

We know that $\det(A^T) = \det(A)$

Since $A^T = -A$,

$\det(A^T) = \det(-A) = (-1)^n \det(A) = \det(A)$

if n is odd, $(-1)^n = -1$,

So we have $-\det(A) = \det(A)$, which indicates that $\det(A) = 0$

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Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\| = \sqrt{\|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2}$

$\|\vec{u}-\vec{v}\| = \sqrt{\|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2}$

if $\|\vec{u}+\vec{v}\| = |\vec{u}-\vec{v}\|$,

we have $\sqrt{\|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2} = \sqrt{\|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2}$, which indicates $2\vec{u}\cdot\vec{v} = -2\vec{u}\cdot\vec{v}$

which indicates that $\vec{u}\cdot\vec{v} = 0$,

which indicates that $\vec{u}$ and $\vec{v}$ are perpendicular